Un-Requested Engine Start

Symptoms

- The engine cranks without the operator pushing the start button on the DCU410 unit or remote panel.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box.	
	STEP 1A. Check the customer interface box logic unit LED illumination.	
	STEP 1B. Check the DCU410 power supply wire for voltage +24-VDC.	
	STEP 1C. Check the remote start supply wire for an open.	
	STEP 1D. Check the starter relay switch signal wire for an open.	
	STEP 1E. Check the SAE J1939 supply and return wires for an open.	
	STEP 1F. Check the SAE J1939 data link supply and return wires for a wire-to-wire short.	

STEPS		SPECIFICATIONS
	STEP 1G. Check the SAE J1939 data link supply wire for a short to ground.	
STEP 2.	Check the OEM wiring harness.	
	STEP 2A. Check the starter relay switch signal and return wires for an open.	
	STEP 2B. Check the starter relay switch signal and return wires for a wire-to-wire short.	
	STEP 2C. Check the starter relay switch signal wire for a short to ground.	

STEP 1. Check the customer interface box.

STEP 1A. Check the customer interface box logic unit LED illumination.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step
Check the crank lamp LED on the DCU 410 unit or remote panel for illumination.	Crank lamp illuminated? **YES**	**1B**
	Crank lamp illuminated? **NO**	**Contact a Cummins® Authorized Repair Location**

STEP 1B. Check the DCU410 power supply wire for voltage +24-VDC.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step

Check the voltage at the battery 1 voltage (switched power) at the DCU410 unit. - Place one test on the battery 1 voltage (switched power) supply wire at the DCU410 unit. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +24-VDC? **YES** **Repair :** Check the batteries. Refer to the OEM service manual.	**Repair complete**
	Less than +24-VDC? **NO**	**1C**

STEP 1C. Check the remote start supply wire for an open.

Conditions :

- Open the customer interface box.
- Disconnect the remote start supply wire from the DCU410 unit.

Action	Specification/Repair	Next Step
Check the remote start supply wire for an open. - Place one test lead on the remote start supply wire at the DCU410 unit. - Place the other test lead remote start supply wire at the X4 connection. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**1D**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 1D. Check the starter relay switch signal wire at the DCU410 unit and C1 connector for an open.

Conditions :

- Open the customer interface box.
- Disconnect the starter relay switch signal wire at the DCU410 unit and C1 connector.

Action	Specification/Repair	Next Step
Check the starter relay switch signal wire at the DCU410 unit and C1 connector for an open. • Place one test lead on the starter relay switch signal wire at the DCU410 unit. • Place the other test lead on the starter relay switch signal wire at the C1 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**1E**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 1E. Check the SAE J1939 data link supply and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the SAE J1939 data link supply and return wires from the DCU410 unit, C3 connector, and X4 connection.

Action	Specification/Repair	Next Step

Check the SAE J1939 data link supply and return wires at the DCU410 unit, C3 connector, and X4 connection for an open. • Place one test lead on the SAE J1939 data link supply wire at the DCU410 unit. • Place the other test lead on the SAE J1939 data link supply wire at the C3 connector. • Place one test lead on the SAE J1939 data link supply wire at the DCU410 unit. • Place the other test lead on the SAE J1939 data link supply wire at the X4 connector. • Place one test lead on the SAE J1939 data link return wire at the DCU410 unit. • Place the other test lead on the SAE J1939 data link return wire at the C3 connector. • Place one test lead on the SAE J1939 data link return wire at the DCU410 unit. • Place the other test lead on the SAE J1939 data link return wire at the X4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**1F**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 1F. Check the SAE J1939 data link supply and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the SAE J1939 data link supply and return wires from the DCU410 unit, C3 connector, and X4 connection.

Action	Specification/Repair	Next Step
Check the SAE J1939 data link supply and return wires at the DCU410 unit for a wire-to-wire short. • Place one test lead on the SAE J1939 data link supply wire at the DCU410 unit. • Place the other test lead on all other wires at the DCU410 unit. • Place one test lead on the SAE J1939 data link return wire at the DCU410 unit. • Place the other test lead on all other wires at the DCU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	**Repair complete**
	Less than 10 ohms? **NO**	**1G**

STEP 1G. Check the SAE J1939 data link supply wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the SAE J1939 data link supply wire at the DCU410 unit.

Action	Specification/Repair	Next Step

Check the SAE J1939 data link supply wire at the DCU410 unit for a short to ground. • Place one test lead on the SAE J1939 data link supply wire at the DCU410 unit. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	**Repair complete**
	Less than 10 ohms? **NO**	**2A**

STEP 2. Check the OEM wiring harness.

STEP 2A. Check the starter relay switch signal and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the starter relay switch signal and return wires at the C1 connector and start motor ring terminal.

Action	Specification/Repair	Next Step

Check the starter relay switch signal and return wires at the C1 connector for an open. • Place one test lead on the starter relay switch signal wire at the C1 connector. • Place the other test lead on the starter relay switch signal wire at the starting motor ring terminal. • Place one test lead on the starter relay switch return wire at the C1 connector. • Place the other test lead on the starter relay switch return wire at the starting motor ring terminal. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**2B**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to the OEM service manual.	**Repair complete**

STEP 2B. Check the starter relay switch signal and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step

Check the starter relay switch signal and return wires at the C1 connector for a wire-to-wire short. • Place one test lead on the starter relay switch signal wire at the C1 connector. • Place the other test lead on all other wires at the C1 connector. • Place one test lead on the starter relay switch return wire at the C1 connector. • Place the other test lead on all other wires at the C1 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to the OEM service manual.	**Repair complete**
	Less than 10 ohms? **NO**	**2C**

STEP 2C. Check the starter relay switch signal wire for a short to ground.

Conditions :
- Open the customer interface box.

Action	Specification/Repair	Next Step

Check the starter relay switch signal wire at the C1 connector for a short to ground. • Place one test lead on the starter relay switch signal wire at the C1 connector. • Place the other test lead on engine ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to the OEM service manual.	**Repair complete**
	Less than 10 ohms? **NO** **Repair :** Replace the connector. Refer to the OEM service manual or contact a Cummins® Authorized Repair Location.	**Repair complete**

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